FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
FULL TEST – IX
PAPER –2
TEST DATE: 12-05-2024

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 51 questions.
- Each subject (PCM) has 17 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section-B & Section-C**.
  Section – A *(01 – 04, 18 – 21, 35 – 38)*: This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
  Section – A *(05 –07, 22 – 24, 39 – 41)*: This section contains **NINE (09)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
  Section – B *(08 – 13, 25 – 30, 42 – 47)*: This section contains **EIGHTEEN (18)** numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.
  Section – C *(14 –17, 31 – 34, 48 – 51)*: This section contains **SIX (06) paragraphs.** Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX).** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	0	In all other cases.

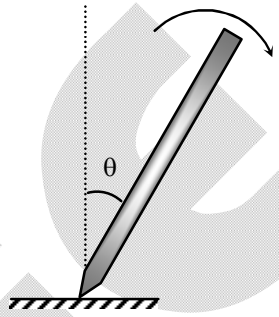
Physics

PART – I

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

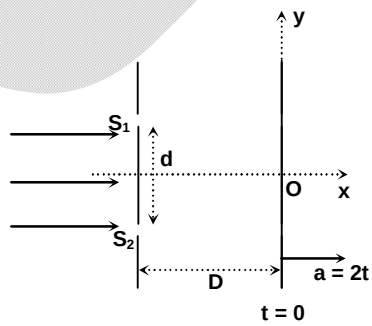
1. A pencil of mass m and length ℓ is placed vertically on the table with its pointed end downwards as shown in the figure. It is then released and tumbles over. If there is sufficient friction to avoid slipping, then at what angle form the vertical the normal force exerted by the pencil becomes $\frac{4}{9}mg$.



(A) $\cos^{-1}\left(\frac{1}{3}\right)$ (B) $\cos^{-1}\left(\frac{2}{3}\right)$

(C) $\cos^{-1}\left(\frac{7}{9}\right)$ (D) $\cos^{-1}\left(\frac{5}{6}\right)$

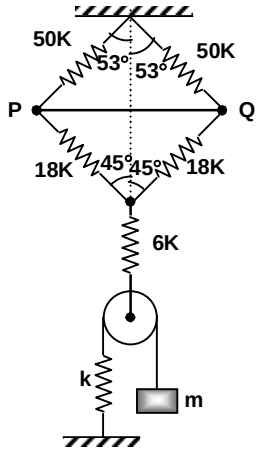
2. Consider the YDSE (young's double slits experiment) arrangement shown in figure. The slits S_1 and S_2 are fixed while the screen in the arrangement starts accelerating from rest in positive x-direction from origin, with acceleration a = 2t m/s^2, at t = 0. Magnitude of velocity of 3rd maximum above the point 'O' at time t = 4 sec in the ground frame is (D >> d and d = 5λ where λ is the wavelength of the light, take $\sqrt{34} = \frac{93}{16}$)



(A) 20 m/s (B) 18.6 m/s

(C) 15 m/s (D) 10 m/s

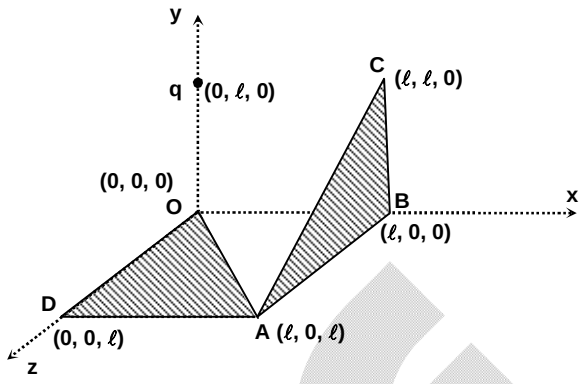
3. System is in equilibrium, rod PQ and pulley are massless as shown in the figure. Find the time period of small vertical oscillations of the block.



(A) $\pi\sqrt{\frac{2m}{k}}$ (B) $2\pi\sqrt{\frac{2m}{k}}$

(C) $\pi\sqrt{\frac{m}{k}}$ (D) $2\pi\sqrt{\frac{m}{2k}}$

4. A point charge 'q' is placed at a distance 'ℓ' on the y-axis. Two shaded triangular portion ABC and AOD lies in the plane y-z and x-z respectively as shown in the figure. The flux of electric field passing through the shaded portion is



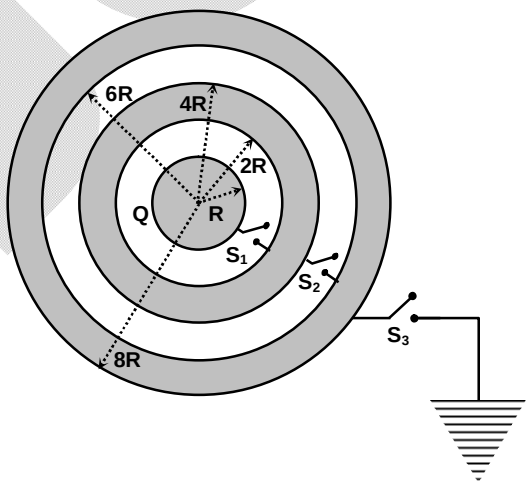
(A) $\frac{q}{4\varepsilon_0}$ (B) $\frac{q}{12\varepsilon_0}$

(C) $\frac{q}{24\varepsilon_0}$ (D) $\frac{q}{36\varepsilon_0}$

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

5. Centre of a conducting solid sphere of radius R having charge Q lies at the centre of two concentric conducting hollow spheres of inner radii 2R and 6R and outer radii 4R and 8R respectively. Then which of the following is/are correct.



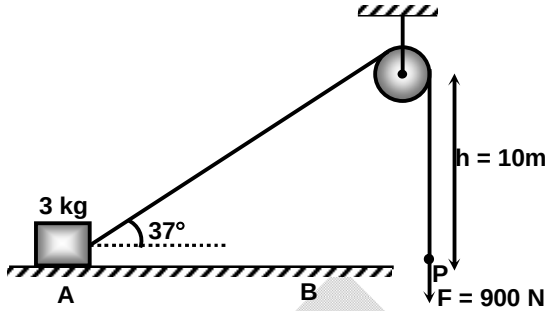
(A) Heat rejected by the system when S_1 is closed while S_2 and S_3 are opened, is $\frac{kQ^2}{4R}$

(B) Heat rejected by the system when S_2 is closed while S_1 and S_3 are opened, is $\frac{kQ^2}{24R}$

(C) Heat rejected by the system when S_1 and S_3 are closed while S_2 is opened, is $\frac{5kQ^2}{16R}$

(D) Heat rejected by the system when S_1, S_2 and S_3 are closed, is $\frac{17kQ^2}{48R}$

6. A block of mass 3 kg is dragged across the smooth floor by the constant force F = 900 N acting at the end of the rope as shown in the figure. Assume that block does not lift off the floor. Then which of the following is/are correct.



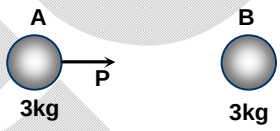
(A) Work done on the block by the force 'F' when the block moves from A to B (at B angle made by the string with the horizontal becomes 53°) is 3750J.

(B) Work done on the block by the force 'F' when the block moves from A to B (at B angle made by the string with the horizontal becomes 53°) is 3250J.

(C) Power delivered by the applied force when the string makes 53° angle with the horizontal is 27000 Watt.

(D) Velocity of the point P when the string makes an angle of 53° with the horizontal, is 30 m/s.

7. Ball 'A' is in motion with linear momentum $30\sqrt{3}$ kg-m/s while ball 'B' is at rest. Masses of both the balls are same and equal to 3 kg. Ball 'A' obliquely collide with the ball 'B'. After the collision, ball 'A' moves at an angle of 30° with the initial direction while momentum of the ball 'B' is $\frac{2}{\sqrt{3}}$ times of its minimum possible value. Then choose the correct option(s).



(A) After the collision both ball 'A' and 'B' moves at an angle of 60° with each other.

(B) After the collision both ball 'A' and 'B' moves at an angle of 90° with each other.

(C) After the collision momentum of the ball 'A' is 30 kg-m/s.

(D) Loss in kinetic energy after the collision is 150 J.

SECTION – B

(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

8. Small amount of a radioactive substance (half life = 30 days) is spread inside a room and consequently the level of radiation becomes 80 times the level for normal occupancy of the room. After how many minimum days will room be safe for occupation?
(Given log 5 = 0.7 and log 2 = 0.3)

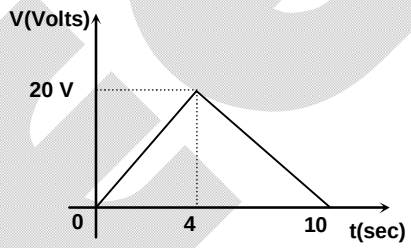
9. There are two vessels each of them contains one mole of a diatomic ideal gas. Initial volume of the gas in each vessel is 12.5×10^{-3} m^3 at 127°C. Equal amount of heat is supplied to each vessel. In one of the vessel, the volume of the gas is doubled without change in its internal energy, where as the volume of the gas is held constant in the second vessel. The vessels are now connected to allow free mixing of the gas. The final temperature of the combined gas is $\frac{\lambda}{10}$ Kelvin. Find the value of λ. (take $\ell n2 = 0.69$)

10. A particle of mass 2 kg is moving in x-y plane according to the relation $y = \frac{x^2}{4}$ with a constant speed 25 m/s. The force acting on the particle at $x = \frac{3}{2}$m normal to the trajectory followed by the particle is λ Newton. Find the value of λ. (x and y are in meter, neglect gravity)

11. A uniform conducting rod of mass m = 20 kg and length ℓ = 2 m is placed on horizontal surface and a uniform magnetic field B = 4 Tesla exist perpendicular to horizontal surface. Suddenly, a certain amount of charge is passed through it, then it is found that rod moves a distance of 5 m on the horizontal surface. The amount of charge that passes through the conductor is $\frac{\lambda}{10}$ coulomb. Find the value of λ. (take g = 10 m/s^2 and coefficient of friction between the horizontal surface on the rod is μ = 1/4)

12. A particle moves along a circular path of radius 6 m about a stationary axis according to the law $\theta = \left(t^3 - 3t^2\right)\pi$ radian. The magnitude of average speed over the time interval between t = 1 to t = 4 sec is $\frac{8\lambda}{100}$ m/sec. Find the value of λ. (given π = 3.14)

13. The potential difference across a 20 H inductor as a function of time is shown in the figure. At t = 0, current in the inductor is zero. The current in the inductor at t = 7 sec is $\frac{P}{100}$ amp. Find the value of P.

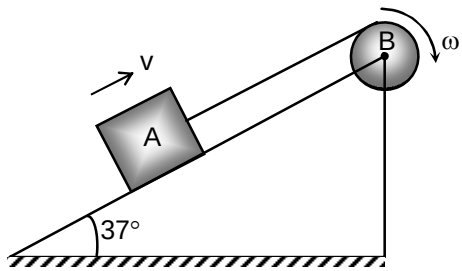


SECTION – C
(Numerical Answer Type)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX).** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Paragraph for Question Nos. 14 and 15

A block A of mass 5 kg is held by a long massless string on a frictional fixed inclined plane of inclination 37° to the horizontal. The string is wound on a uniform solid cylindrical drum B of mass 10 kg and of radius 2 m. Initially cylinder is given an angular velocity as shown in the figure. Then answer the following questions. (Take g = 10 m/s^2, tan 37° = $\frac{3}{4}$ and the coefficient of friction between the block and the inclined plane is $\mu = \frac{1}{2}$)



14. The tension in the string during the motion is ……Newton.

15. At a certain instant of time the magnitude of the angular velocity of cylinder is 10 rad/s, then the distance travelled by the block A from that instant of time until it comes to rest is …….meter.

Paragraph for Question Nos. 16 and 17

A vernier callipers is manufactured in which the main scale of vernier callipers reads in millimetre and its vernier is divided into 8 divisions, which coincide with 5 divisions of main scale. Then answer the following questions.

16. When the two jaws of the instrument touch each other the zero of the vernier coincide with the zero of the main scale. A rod is tightly placed along its length between both jaws, it is observed that the zero vernier scale lies just right to 38th division of main scale and seventh division of vernier scale coincide with the main scale. The measured value is $\frac{P}{100}$ cm. The value of P is

17. If when the two jaws of instrument touch each other the 9th division of vernier scale coincide with one of the main scale division of main scale and zero of vernier scale is slightly right to the zero of main scale. Then zero error is positive zero error of magnitude $\frac{Q}{10}$ mm. The value of Q is

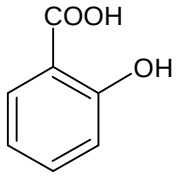
Chemistry

PART – II

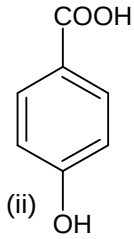
SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

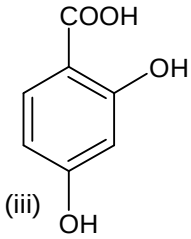
18. The decreasing order of pK_a values for the following acid is



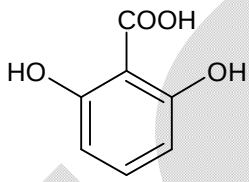
(i)



(ii)



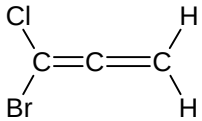
(iii)

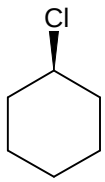


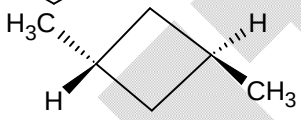
(iv)

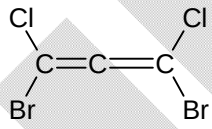
(A) iv > i > iii > ii
(B) ii > iv > i > ii
(C) ii > i > iii > iv
(D) ii > iii > i > iv

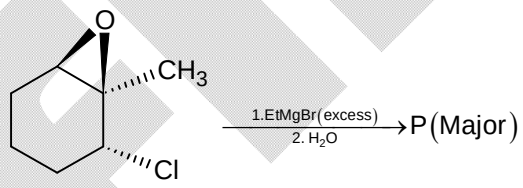
19. Optically active compound is/are:

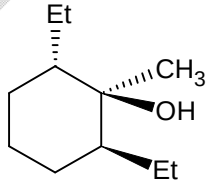
(A) 

(B) 

(C) 

(D) 

20. 

Choose the correct option is/are
(A) 'P' is optically inactive
(B) 'P' is optically active
(C) 'P' has 2 chiral centres
(D) 'P' is 

21. 4 mole of a mixture O_2 and O_3 is reacted with excess of acidified solution of KI. The liberated iodine require 1 L of 2 M hyposolution for complete reaction. The mole% of O_3 in the initial sample is
(A) 25 (B) 60
(C) 75 (D) 50

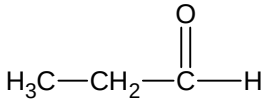
SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

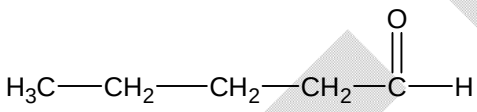
22. High spin complex is/are
(A) $\left[Fe(H_2O)_6\right]^{2+}$ (B) $\left[Sc(H_2O)_6\right]^{3+}$
(C) $\left[Co(H_2O)_6\right]^{2+}$ (D) $\left[Co(C_2O_4)_3\right]^{3-}$

23. Which of the following is/are homo polymer
(A) Butadiene – styrene (Buna-S) (B) Polypropene
(C) Polyethene (D) Nylon-6

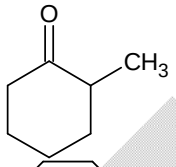
24. Tautomer of which of the following can show geometrical isomerism?
(A)



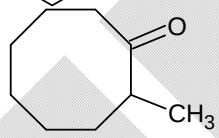
(B)



(C)



(D)

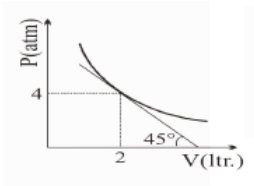


SECTION – B
(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

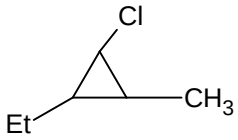
25. On passing electricity through nitrobenzene solution, it is converted into azobenzene. The value of mass of azobenzene produced (in gram), if the same quantity of electricity produces oxygen just sufficient to burn 96 g of fullerene (C_{60}) is [molar mass of azobenzene = 182 g/mol]

26. The heat capacity of the monoatomic ideal gas undergoing the process, as shown in the P-V graph is xR, the value of 2x is (Where R is gas constant)

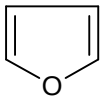


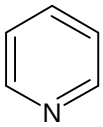
27. 200 ml of a saturated solution of $CaC_2O_4(s)$ requires 8 ml of $5\times10^{-3}[N]\,KMnO_4$ solution of complete oxidation in the presence of acid. The value of $-\log K_{SP}$ saturated solution of $CaC_2O_4(s)$ [Nearest possible integer]

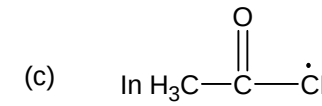
28. $K_2Cr_2O_7 + xHCl \longrightarrow 2KCl + yCl_2 + zCrCl_3 + pH_2O$
The value of x is

29. Total number of diastereomeric pair for  is x, value of x is

30. Total number of correct statements among the following is/are

(a)  is aromatic.

(b)  is more reactive than benzene for EAS reaction.

(c) In  , the presence of  group stabilize the radical.

(d) $[\Delta S]_{system}$ for reversible adiabatic process = 0.

(e) Boiling point and freezing point are intensive property.

SECTION – C
(Numerical Answer Type)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX).** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Paragraph for Question Nos. 31 and 32

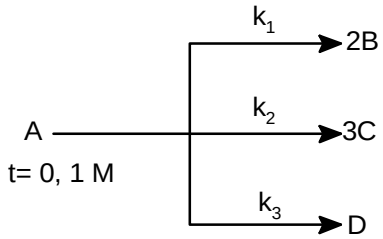
A closed rigid bulb contains 0.01 mole of inert helium gas and a sample of solid white NH_4Cl. Assume that the volume of solid NH_4Cl is negligible compared to volume of the bulb. The pressure of He is measured at 300 K and found to be 114 mm Hg. The bulb is then heated to 600 K. All the NH_4Cl decomposes to $NH_3(g)$ and HCl(g). The final total pressure in the bulb after complete decomposition of the solid is 908 mm Hg. Assume ideal behavour for all gas

31. The mass of NH_4Cl present initially in the bulb is y. Value of 2y is

32. Volume of bulb is

Paragraph for Question Nos. 33 and 34

For 1st order parallel reaction given below $k_1 = 0.1\,min^{-1}, k_2 = k_3 = 0.2\,min^{-1}$ initial concentration of A is 1 M.



33. The value of concentration "B" at t = 4.606 M

34. The value of concentration "C" at t = 4.606 min is

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

35. Let α, β be the roots of $ax^2 + bx + c = 0$. Let $S_n = \alpha^n + \beta^n$, $n \geq 1$ and $\Delta = \begin{vmatrix} 1+S_0 & 1+S_1 & 1+S_2 \\ 1+S_1 & 1+S_2 & 1+S_3 \\ 1+S_2 & 1+S_3 & 1+S_4 \end{vmatrix}$. If α, β are distinct and real, then

(A) $\Delta \leq 0$ (B) $\Delta > 0$

(C) $\Delta < 0$ (D) $\Delta = 0$

36. If $(1 + x)^n = C_0 + C_1x + ... + C_nx^n$, then the value of $\sum\limits_{0\leq r<s\leq n}\sum C_rC_s$ is equal to

(A) $\frac{1}{2}\left[2^{2n} - {}^{2n}C_n\right]$ (B) $\frac{1}{4}\left[2^{2n} - {}^{2n}C_n\right]$

(C) $\frac{1}{2}\left[2^{2n} + {}^{2n}C_n\right]$ (D) $\frac{1}{2}\left[2^{n} - {}^{2n}C_n\right]$

37. Let f(x) be a derivable function at x = 2 and $\lim\limits_{h\to 0}\frac{f(2+h)}{\sin h} = 3$, then $\left|\frac{f(2)+f'(2)}{f'(2)-f(2)}\right|$ is

(A) 1 (B) 0

(C) 2 (D) 3

38. If $f(x) = x^2 + ax + b$ where a, b $\in$ R and f(f(x)) = 0 has roots 1 and 2. If k be the value of f(x) at x = 0, then k is

(A) $\frac{3}{2}$ (B) 2

(C) $-\frac{3}{2}$ (D) –2

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

39. If PA and PB are tangents drawn to a circle whose centre is O where A and B are points of contact such that $\frac{1}{(OA)^2} + \frac{1}{(PA)^2} = \frac{1}{16}$, then

(A) AB = 8 units

(B) AB = 4 units

(C) area of quadrilateral OAPB is equal to 4(OP) sq. units

(D) area of quadrilateral OAPB is equal to 8(OP) sq. units

40. There are three coins C_1, C_2 and C_3, C_1 is a fair coin painted blue on the head side and white on the tail side. C_2 and C_3 are biased coins so that the probability of a head is p. They are painted blue on the tail side and red on the head side. Two of the three coins are selected at random and tossed. If the probability that both the coins land up with sides of the same colour is $\frac{29}{96}$, then the possible value(s) of p can be

(A) $\frac{1}{8}$ (B) $\frac{3}{8}$

(C) $\frac{5}{8}$ (D) $\frac{7}{8}$

41. Number of ways in which 32 identical blankets can be distributed among 4 persons so that each of them gets number of blankets which are divisible by 2 but not by 4, is/are

(A) ${}^9C_3$ (B) ${}^{19}C_3$

(C) ${}^9C_6$ (D) ${}^4C_1 + 5\cdot {}^4C_2 + 10\cdot {}^4C_3 + 10\cdot {}^4C_4$

SECTION – B

(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

42. Let M be a 2×2 matrix such that $M\begin{bmatrix}1\\-1\end{bmatrix} = \begin{bmatrix}-1\\2\end{bmatrix}$ and $M^2\begin{bmatrix}1\\-1\end{bmatrix} = \begin{bmatrix}1\\0\end{bmatrix}$. If x_1 and $x_2 (x_1 > x_2)$ are the two values of x for which det (M – xI) = 0, where I is an identity matrix of order 2, then find the value of $(5x_1 + 2x_2)$

43. Tangents are drawn from the point (–1, 2) on the parabola $y^2 = 4x$. The length, the tangents will intercept on the line x = 2 is k, $\frac{k}{\sqrt{2}}$ is equal to

44. If $\tan^{-1}\left(x+\frac{3}{x}\right) - \tan^{-1}\left(x-\frac{3}{x}\right) = \tan^{-1}\frac{6}{x}$, then the value of $5x^8 - 4x^4 + 7$ equals

45. If α and β are roots of the equation $\begin{bmatrix}1 & 25\end{bmatrix}\begin{bmatrix}0 & \frac{1}{2}\\ -1 & \frac{1}{2}\end{bmatrix}^5\begin{bmatrix}1 & -1\\2 & 0\end{bmatrix}^{10}\begin{bmatrix}0 & \frac{1}{2}\\ -1 & \frac{1}{2}\end{bmatrix}^5\begin{bmatrix}x^2-5x+20\\ x+2\end{bmatrix} = [40]$, then find the value of $(1-\alpha)(1-\beta)$

46. Find the shortest distance between the curves $y^2 = x^3$ and $9x^2 + 9y^2 - 30y + 16 = 0$ is k, then 10k is equal to

47. If $\left(\frac{a^3}{a-1}, \frac{a^2-3}{a-1}\right), \left(\frac{b^3}{b-1}, \frac{b^2-3}{b-1}\right)$ and $\left(\frac{c^3}{c-1}, \frac{c^2-3}{c-1}\right)$ are collinear and $\alpha abc + \beta(a+b+c) = \gamma(ab+bc+ca)$, $\alpha, \beta, \gamma \in N$, then find the least value of $\alpha + \beta + \gamma$

SECTION – C
(Numerical Answer Type)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX).** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Paragraph for Question Nos. 48 and 49

Let a real valued function f defined on real number satisfies $\int_0^{2x} t^2 f(2x-t)\,dt = 2e^{2x} - 4x^2 + 2x - 2$ and

$g(x) + g'(x) = f(-x)$ with $g(1) = \frac{1}{e}$

48. $f^{-1}(e)$ is equal to

49. Area bounded by $y = g(x)$ and x-axis will be

Paragraph for Question Nos. 50 and 51

Let A and B be points on the same branch of the hyperbola xy = 1. Suppose that P is a point lying between A and B on this hyperbola, such that the area of the triangle APB is maximum

50. Let $A \equiv (1, 1)$ and $B \equiv \left(4, \frac{1}{4}\right)$, then abscissa of P is equal to

51. $\frac{\text{Area of the region bounded by the hyperbola and chord AP}}{\text{Area of the region bounded by the hyperbola and chord PB}}$ is equal to